



Fluorescence Dissolved Oxygen Transmitter

User Manual

PR-3001-LDO-N01

Ver 1.0



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No.1Product Introduction

1.1Product Overview

This product is a device for measuring the dissolved oxygen concentration of a solution. It uses the fluorescence measurement principle, does not consume oxygen, and does not require electrolyte. It has a built-in temperature transmitter with automatic temperature compensation function. The calibration function provides a simpler and more convenient calibration method. It can be widely used in water treatment, aquaculture, environmental monitoring and other industries.

1.2Features

- Dissolved oxygen measurement range 0~20mg/L (0~200%saturation).
- RS485 Communication interface: MODBUS RTU The communication protocol can be easily connected to the computer for monitoring and communication

News.

- ModBus The communication address can be set and the baud rate can be modified.
- The equipment adopts wide voltage power supply, DC10~30V Both are OK. It uses the fluorescence
- measurement principle, does not consume oxygen, and does not require electrolyte.

1.3Main parameters

powered by	DC 10~30V
Power consumption	0.2W
Communication interface	RS485; Standard MODBUS-RTU Protocol; Communication baud rate: default 4800 (2400, 4800, 9600 (Can be set))
Measuring principle	Fluorescence method
Measuring range	0~20mg/L (0~200% Saturation)
Measurement error	$\pm 3\%FS \pm 0.5^{\circ}C$ (25 $^{\circ}C$)
Resolution	0.01mg/L; 0.1%; 0.1 $^{\circ}C$
Response time	≤ 60 sec
Equipment working conditions	0~40 $^{\circ}C$
Fluorescent film life	Normal use 1 Year
Storage conditions	- 10~60 $^{\circ}C$
Waterproof grade	IP68
Electrode wire length	default 5m
Shell material	Corrosion-resistant plastic, stainless steel

1.4Shell appearance and size



1.5 Product Selection

PR-				Company Code
	3001-			First generation shell
		LDO-		Fluorescence Dissolved Oxygen Transmitter
			N01	RS485 (Modbus-RTU protocol)

No.2Chapter Hardware Connection

2.1Equipment pre-installation inspection

Equipment List:

- ◆ Fluorescence dissolved oxygen transmitter 1 set
- ◆ 5m cable
- ◆ Certificate of Compliance

2.2Interface Description

The power interface is a wide voltage power input10-30VBoth are acceptable.485Pay attention when wiring the signal lineA\BThe two wires cannot be connected reversely, and the addresses of multiple devices on the bus cannot conflict.

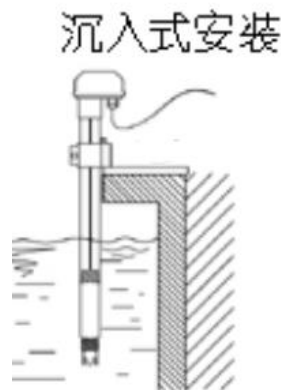
2.2.1Sensor Wiring

	illustrate	illustrate
electricity source	brown	Power supply positive (10~30V DC)
	black	Negative power supply
Pass News	green	485-A
	blue	485-B

2.3Installation

The sensor should be immersed below the liquid surface for fixed installation. Avoid collision or scratching the surface of the fluorescent membrane head during installation and use. The fluorescent membrane head should be prevented from being attached by sediment at the bottom of the water. The rubber protective cover should be removed during use. Submerged installation:

With3/4The thread can be used with our waterproof pipe. The cable passes through the pipe and the device is screwed into the thread of the waterproof pipe.



No.3Chapter Configuration Software Installation and Usage

Our company provides supporting "485"Parameter Configuration Software" can easily use the computer to read the sensor parameters and flexibly modify the sensor equipmentIDand address.

Note that when using software to automatically obtain485There is only one sensor on the bus.

3.1Parameter configuration description

Open the data package and select "Debug Software"----"485Parameter Configuration Software", find it and open it.

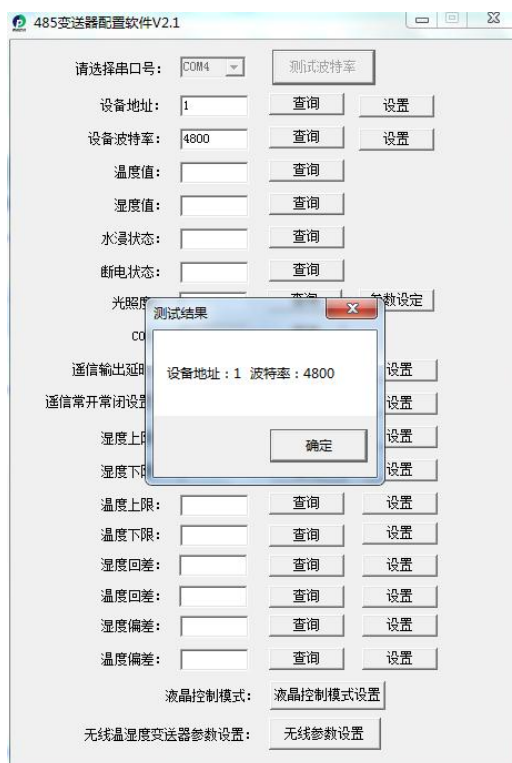


1) Choose the correctCOMPort (check in "My Computer - Properties - Device Manager - Ports"COM port), the following figure lists several different485The name of the converter driver.



2) Connect only one device and power it on. Click the test baud rate in the software. The software will test the baud rate and address of the current device. The default baud rate is4800bit/s,The default address is0x01.

3) Modify the address and baud rate according to the needs, and query the current functional status of the device. 4) If the test is unsuccessful, recheck the device wiring and485Driver installation status.



3.2Calibration Instructions

3.2.1 Calibration Preparation

Zero oxygen standard solution preparation: Preparation 1 beaker, take 200mL Pour distilled water into the cup, then add anhydrous sodium sulfite, stirring while adding, until the anhydrous sodium sulfite is saturated, the standard solution is close to 0 oxygen.

100% Oxygen environment preparation: Preparation 1 beaker, take 200mL Pour pure water (or distilled water) into the cup, add an air pump, and fully expose the solution to air (at least 30 min). Note: If the on-site conditions do not allow or the use requirements are not high, the sensor can also be placed directly in the air (the calibration accuracy will deviate).

3.2.2 calibration

In the Configuration Tool, find the Dissolved Oxygen tab in the Water Quality Sensors tab. Place the sensor in 100% In an oxygen environment, after the dissolved oxygen value stabilizes, click the full scale calibration button to complete 100% Oxygen saturation calibration.

Then put the sensor into the zero oxygen standard solution, stir it gently to speed up the response, and then let it stand. After the dissolved oxygen value stabilizes, click the zero oxygen calibration button to complete the zero oxygen calibration.

No.4 Chapter Communication Protocol

4.1 Basic communication parameters

coding	8bit binary
Data bits	8Bit
Parity bit	none
Stop bits	1Bit
Error checking	CRC(Redundant Cyclic Code)
Baud rate	2400bit/s, 4800bit/s, 9600 bit/s Can be set, factory default is 4800bit/s

4.2 Data frame format definition

use Modbus-RTU Communication protocol, the format is as

follows: Initial structure ≥ 4 Bytes of time

Address code = 1 byte

Function code = 1 byte

Data Area = N byte

Error Check = 16 Bit CRC Code

end structure ≥ 4 Bytes of time

Address code: The address of the transmitter, which is unique in the communication network (factory default 0x01). Function code:

Function indication of the command sent by the host.

Data area: The data area is the specific communication data.16 bitsThe data high byte comes

first! CRCCode: Two-byte check code.

4.3 Register Address

Register Address	operate	illustrate
0000H,0001H	03	Dissolved oxygen saturation (%; large end of floating point number)
0002H,0003H	03	Dissolved oxygen concentration (mg/L; floating point number)
0004H,0005H	03	Temperature (°C; floating point big endian)
1010H	06	Calibration (write 0x0001 to calibrate zero point, Write 0x0002 to calibrate 100% saturation point)
1020H	03/06	Salinity (‰; default 0)
1022H	03/06	Atmospheric pressure (kPa; default 101.33, actual value magnified 100 times)

4.4 Communication protocol examples and explanations

4.4.1 The read address is 01 The current dissolved oxygen saturation (%), dissolved oxygen concentration (mg/L)

and temperature

Send frame:

Address code	Function code	Register Address	Register contents	Check code low	Check code high digit
0x01	0x03	0x00 0x00	0x00 0x06	0xc5	0xc8

Response frame:

Address code	Function code	Number of valid bytes	Register contents	Check code low	Check code high digit
0x01	0x03	0x0c	0x3f 0x6a 0xeb 0x52 0x40 0xe2 0x48 0xb 0 0x41 0xe5 0x85 0x c5	0xa7	0x49

Floating point big endian 3f6aeb52 for 0.917653 The dissolved oxygen saturation is 91.8% Floating

point big endian 40 e2 48 b0 for 7.071373 The dissolved oxygen concentration is 7.07mg/L Floating

point big endian 41 e5 85 c5 for 28.690317 Indicates the temperature 28.7°C

4.4.2 Write atmospheric pressure

Send frame: (to address code 1) The device writes the atmospheric pressure

101.35kPa Towards 0x1022 Register Write 10135 Convert to hexadecimal 2797

Address code	Function code	Register Address	Register contents	Check code low	Check code high digit
0x01	0x06	0x10 0x22	0x27 0x97	0x77	0x5e

Response frame: (according to MODBUS) The standard response is a mirrored message of the sent frame)

Address code	Function code	Register Address	Register contents	Check code low	Check code high digit
0x01	0x06	0x10 0x22	0x27 0x97	0x77	0x5e

4.4.3 Zero point calibration

After the sensor is placed in oxygen-free water and the value is stable, 1010H Register Write 0x0001

Address code	Function code	Register Address	Register contents	Check code low	Check code high digit
0x01	0x06	0x10 0x10	0x00 0x01	0x4d	0x0f

Response frame: (according to MODBUS) The standard response is a mirrored message of the sent frame)

Address code	Function code	Register Address	Register contents	Check code low	Check code high digit
0x01	0x06	0x10 0x10	0x00 0x01	0x4d	0x0f

No.5Chapter Precautions and Maintenance

- ◆ When the equipment has obvious faults, please do not open it and repair it by yourself, contact us as soon as possible!
- ◆ Avoid collision or scratching of the fluorescent film on the front of the device. Any damage will lead to reduced measurement accuracy or even make it unusable.
- ◆ Avoid using in organic solvents and avoid using organic solvents to clean the fluorescent cap.
- ◆ When installing the equipment, try to avoid cables being too tight or stressed.
- ◆ Equipment cleaning:

For the outer surface of the equipment, you can use tap water to clean it and wipe it with a damp soft cloth. For some stubborn dirt, you can add some household detergent to the tap water to clean it.

For the outer surface of the fluorescent cap, use clean water to rinse the dirt on the sensor light window; if it needs to be wiped, wipe it gently with a soft cloth and do not scratch it hard to prevent damage to the fluorescent film, resulting in inaccurate measurement and inability to measure.

If dust or water vapor enters the fluorescent cap: unscrew the fluorescent cap, rinse the inner surface of the fluorescent cap and the optical glass window of the device with tap water. If there is grease-containing dirt, wash it with tap water mixed with household detergent, then rinse off the detergent, wipe all cleaned surfaces with a lint-free soft cloth, and then put it in a dry place to allow the water to evaporate completely.